

Comparison of MRI k-Space Sampling Trajectories Using Non-Uniform Fourier Reconstruction

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Abstract - Non-Cartesian magnetic resonance imaging (MRI) sampling trajectories can reduce acquisition time and provide greater flexibility in k-space acquisition, but their irregular sampling patterns introduce additional computational challenges during image reconstruction [1]. This study investigates the performance of several k-space sampling trajectories and evaluates the use of a gridded non-uniform fast Fourier transform (NUFFT) as a computationally efficient alternative to the non-uniform discrete Fourier transform (NUDFT) for MRI reconstruction [2] [3]. A C++ based MRI simulation was developed to generate Cartesian, radial, spiral, random, and variable density sampling trajectories and reconstruct images using non-uniform Fourier methods. Reconstruction quality was evaluated using peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM), while computational efficiency was evaluated by measuring reconstruction time. Additional experiments investigated the effects of reduced sampling, additive noise, and simulated transverse motion on reconstruction quality. The results demonstrate that NUFFT-based reconstruction substantially reduces computation time compared with the direct NUDFT approach, with the greatest efficiency improvements occurring at higher sample counts, while maintaining similar reconstruction quality. Across the trajectory comparisons, Cartesian variable-density sampling generally produced the highest PSNR and SSIM values, while radial and spiral trajectories provided better preservation of fine detail and greater resistance to simulated motion. Spiral sampling provided a favorable overall balance across the conditions tested; however, the most effective trajectory could not be determined based on the experiments conducted. These results demonstrate the potential of the developed simulation as a workflow for evaluating and optimizing k-space sampling trajectories and show that the most effective trajectory depends on the specific imaging conditions and reconstruction objectives.

I. Introduction

Magnetic Resonance Imaging (MRI) acquires images by sampling spatial frequency information, known as *k-space* [4] [5]. Traditional MRI acquisitions typically use Cartesian sampling trajectories, which uniformly sample k-space, and when fully sampled, allow for high-quality image reconstruction [1] [4] [6]. However, full Cartesian sampling requires a large number of measurements, resulting in relatively long acquisition times.

Acquisition time can be reduced by under-sampling k-space. For example, to fully reconstruct a 256×256 image using a Cartesian sampling trajectory requires 65,536 k-space samples. If the image is undersampled using 30,000 samples, it can still be reconstructed in a shorter acquisition time, but the reduced sampling generally leads to image degradation and visual artifacts that may conceal crucial features [1].

Because low spatial frequencies near the center of k-space generally contain significant information about overall image contrast and structure, acquisition efficiency can be improved

by using non-Cartesian sampling trajectories that prioritize these regions [7]. For example, radial trajectories repeatedly pass through the center of k-space, while variable density trajectories fully sample the central frequencies and gradually reduce the sampling density toward the edges [8]. These approaches aim to preserve image quality while reducing acquisition time.

The objective of this study is to compare several k-space sampling trajectories by simulating MRI acquisition and reconstruction using a Non-Uniform Discrete Fourier Transform (NUDFT) [3]. The reconstructed images were evaluated using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM) [9], and computational time. Additionally, the effects of trajectory-specific parameters will be investigated to determine the optimal sampling configurations under different imaging conditions.

II. Theory

The Fourier Transform is a mathematical transform that takes a function (or a signal) and separates it into its frequency components. The transform comes from the Fourier series, which states that any periodic, continuous, or piecewise-smooth function can be represented as an infinite sum of sinusoidal waves of varying frequencies and amplitudes. The Fourier Transform is defined as [10]:

$$F(k) = \int_{-\infty}^{\infty} f(x) e^{-i2\pi kx} dx$$

Where:

- x is the spatial coordinate.
- κ is spatial frequency.

It takes a function $f(x)$ and transforms it into another function $F(\kappa)$ which takes a frequency κ and outputs how much that frequency is contributing to the original function/signal.

The first key idea that makes the Fourier Transform useful is the Inverse Fourier Transform, defined as:

$$f(x) = \int_{-\infty}^{\infty} F(\kappa) e^{i2\pi kx} d\kappa$$

The inverse transform allows for a seamless transition from the spatial/time domain to the frequency domain, and vice versa. The second key idea is that these concepts can be expanded to discrete values. These transforms are known as the

Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT). Defined as [11]:

$$DFT: F(\kappa) = \sum_{n=0}^{N-1} f(n) e^{-i2\pi\kappa n/N}$$

$$IDFT: f(n) = \frac{1}{N} \sum_{\kappa=0}^{N-1} F(\kappa) e^{i2\pi\kappa n/N}$$

Where:

- N is the number of samples within the dataset.
- n is the current index position
- κ is spatial frequency
- $f(n)$ is the value of the dataset f at n
- $F(\kappa)$ is the Fourier Transform of f at n

In the inverse transform, the summation is normalized by the N to distribute the spectral power across the reconstructed samples.

The Fast Fourier Transform (FFT) and its inverse (IFFT) are algorithms used to calculate the DFT and IDFT respectively. They take the order of operations from $O(N^2)$ to $O(N \log_2 N)$, which drastically decreases computational time [12] [13] [14].

The Fourier Transform can be extrapolated to 2 dimensions by applying the transform twice, once in the x direction and once in the y direction. The result is the definition of the 2D Fourier Transforms [10]:

$$2D\ FT: F(\kappa_x, \kappa_y) = \iint f(x, y) e^{-i2\pi(\kappa_x x + \kappa_y y)} dx dy$$

$$2D\ DFT: F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-i2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

Similarly, the inverse transforms are:

$$2D\ IFT: f(x, y) = \iint F(\kappa_x, \kappa_y) e^{i2\pi(\kappa_x x + \kappa_y y)} d\kappa_x d\kappa_y$$

$$2D\ IDFT: f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{i2\pi(\frac{ux}{M} + \frac{vy}{N})}$$

The result of a 2D Fourier Transform is a grid of spatial frequencies composed of κ_x and κ_y , commonly referred to as *k-space*. Generally, frequencies are organized such that towards the center contains lower frequencies, which control the overall brightness and contrast of the image, while the edges contain higher frequencies that control more of the finer details [7].

The 2D Fourier Transform is the concept that enables MRI systems to image cross sections of tissue. MRIs collect k-space samples by applying a constant strong magnetic field, usually 1.5 Tesla or 3.0 Tesla, along with x, y, z gradient coils, all of which align magnetic moments of the hydrogen nuclei with the

direction of the magnetic field. RF pulses are used to excite the nuclei causing the magnetic moments to precess at its Larmor frequency, defined as:

$$\omega = \gamma B_0$$

Where:

- γ = gyromagnetic ratio ($\frac{MHz}{T}$)
- B_0 = base magnetic field (T).

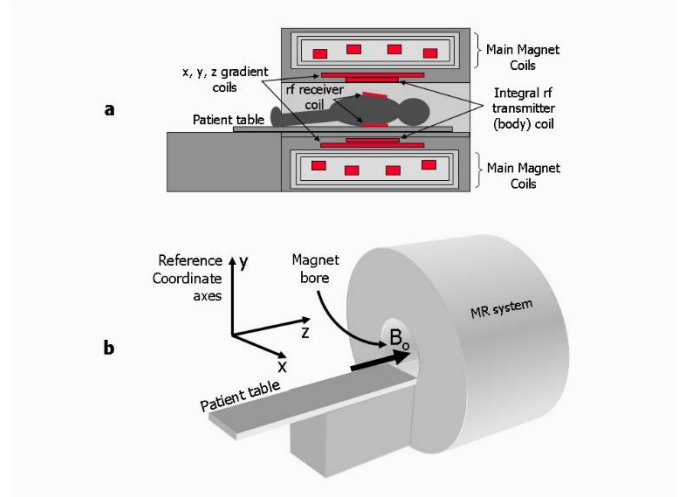


Figure 1. Schematic of an MRI system illustrating the main magnetic field, gradient coils used for spatial encoding, RF transmit and receive coils, and the reference coordinate system. Diagram reproduced from [7].

The excited nuclei gradually relax back toward equilibrium. The resulting transverse magnetization induces a voltage in the receiver coil according to Faraday's law. The received signal is recorded as a complex signal using quadrature detection to separate the signal into its sine and cosine components, forming the complex Fourier coefficients [4] [5] [7].

To encode spatial data within the AC signal, slight gradients in the x, y, and z directions are applied at precise times, making the Larmor Frequency dependent on its spatial coordinates:

$$\omega(x) = \gamma(B_0 + G_x \cdot x)$$

Where:

- ω is Larmor Frequency
- B_0 is base magnetic field (1.5T or 3.0T)
- x is x-coordinate
- G_x is the gradient in X direction ($\frac{T}{m}$)

During the signal readout the nuclei that are located in the high end of the gradient will precess faster than the nuclei in the low end of the gradient, thus gaining more phase. This means that the phase is now a function of space and time, such that:

$$\phi(x, t) = \int_0^t \omega(x) dt = \gamma(B_0 + G_x \cdot x)t$$

Ignoring the constant phase shift caused by B_0 :

$$\begin{aligned} \phi(x, t) &= \gamma G_x x \cdot t \\ \text{let: } k_x &= \gamma G_x t \left(\frac{\text{rad}}{m} \right) \\ \Rightarrow \phi(x, t) &= k_x(t) \cdot x \end{aligned}$$

Through Faraday's Law, the rotating transverse magnetization induces an RF voltage within the receiving coil. Therefore, the voltage signal $S(t)$ at time = t is the sum of all the voltages caused by all the spins, or:

$$\begin{aligned} S(t) &= \int \rho(x) e^{-i\phi(x,t)} dx \\ \Rightarrow S(t) &= \int \rho(x) e^{-i\gamma G_x x t} dx \end{aligned}$$

Where $\rho(x)$ is the spin density. It is essentially the unknown function that describes the distribution of protons through the tissues. Comparing $S(t)$ to the Fourier Transform:

$$S(k_x) = \int \rho(x) e^{-i2\pi k_x x} dx$$

Setting the equations equal gives:

$$\begin{aligned} 2\pi k_x &= \gamma G_x t \\ \Rightarrow k_x &= \frac{\gamma G_x t}{2\pi} \left(\frac{\text{cycles}}{m} \right) \end{aligned}$$

To encode k_y , a pulse of another slight magnetic gradient perpendicular to the first is applied before the next readout. This pulse creates a brief change in the precession frequencies of the hydrogen nuclei. This causes phase to accumulate through the duration of the pulse. The phase accumulated will remain after the pulse, thus creating a gradient of phase shifts for the next readout forming one line of k_y . Such that:

$$\begin{aligned} \phi(y) &= \gamma G_y y \Delta t \\ \text{Thus: } k_y &= \gamma G_y \Delta t \end{aligned}$$

For a line of k_y in k-space, the amplitude of G_y will increase and the process will repeat until the last sample. Expanding the previous integral:

$$\begin{aligned} S(t) &= \iint \rho(x, y) e^{-i\phi(x,y,t)} dx dy \\ \phi(x, y, t) &= (\omega_x \cdot T_p) + (\omega_x \cdot t) \end{aligned}$$

Where:

- T_p is Pulse Duration

In practice, MRI gradients vary throughout the pulse sequence. Therefore, the more general definition is:

$$\begin{aligned} \kappa_x &= \frac{\gamma}{2\pi} \int G_x(t) dt \\ \kappa_y &= \frac{\gamma}{2\pi} \int G_y(t) dt \end{aligned}$$

Which changes the full phase equation, ignoring the constant γB_0 phase-shift, to be defined as:

$$\phi(x, y, t) = \gamma x \int_0^t G_x(t) dt + \gamma y \int_0^t G_y(t) dt$$

Comparing the MRI equation with the 2D Fourier Transform shows that the measured MRI signal is mathematically equivalent to the Fourier transform when considering all readouts. Consequently, MRI acquisition can be interpreted as a physical implementation of the Fourier transform [1] [4] [5] [15].

$$\text{MRI Equation: } S(t) = \iint \rho(x, y) e^{-i\phi(x,y,t)} dx dy$$

$$2D \text{ FT: } S(k_x, k_y) = \iint \rho(x, y) e^{-i2\pi(k_x x + k_y y)} dx dy$$

This paper will be looking at k-space acquisition trajectories. Conventional MRI systems typically acquire k-space using a Cartesian trajectory (row by row). However, research has investigated new non-Cartesian trajectories.

Because low spatial frequencies near the center of k-space contain much of the information associated with overall image intensity and contrast, new trajectories allow for the most important frequencies to be acquired first, which could potentially lead to shorter acquisition times while maintaining adequate definition. MRI systems follow different k-space trajectories by applying time-varying magnetic field gradients in the x and y directions [1] [7].

The time varying gradients are applied to determine the k-space coordinates, allowing for Cartesian lines, radial spokes, spiral paths, or other sampling patterns [1]. This project compares several k-space sampling trajectories using custom-made simulated MRI acquisitions to evaluate the effects on reconstruction quality and computational performance.

III. Methods

**The source code and simulation framework developed for this study are publicly available at: https://github.com/UnderscoreLogs/MRI_k-Space_Acquisition_NUFT_Simulator*

The main drawback of the traditional FFT is that it assumes uniformly spaced samples on a Cartesian grid. Consequently, the FFT cannot be directly applied to data acquired using non-Cartesian sampling trajectories [1]. To solve this problem, this project uses a direct non-uniform discrete Fourier transform (NUDFT) reconstruction with density compensation weighting (DCF) [2] [3].

The NUDFT implemented in this program was derived from first principles of MRI systems. The algorithm works by treating every pixel's brightness as the spin density and then summing all pixels to form one k-space sample.

The forward NUDFT is defined as [3]:

$$S(\kappa_x, \kappa_y) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} I(x, y) e^{-i2\pi(\kappa_x X(x) + \kappa_y Y(y))}$$

where:

- $S(\kappa_x, \kappa_y)$ is the acquired k-space sample at spatial frequency coordinates (κ_x, κ_y)
- M is the image width in pixels
- N is the image height in pixels
- $I(x, y)$ is the image intensity at pixel (x, y)
- $X(x)$ and $Y(y)$ are the encoded spatial coordinates of the pixel
- κ_x and κ_y define the sampled k-space trajectory location

Unlike the conventional DFT, this NUDFT allows for (κ_x, κ_y) to be sampled arbitrarily positioned in k-space, which enables reconstruction from non-Cartesian trajectories [1].

The image is reconstructed by applying the inverse Fourier transform to the acquired non-uniform k-space samples. Since non-Cartesian trajectories produce uneven sampling density, a density compensation factor (DCF) is added to correct for the k-space sampling density depending on the sampling trajectory. Similarly, the inverse NUDFT (INUDFT) is defined as [3] [16]:

$$I(x, y) = \sum_{i=0}^{O-1} DCF_i S_i e^{i2\pi(\kappa_x X(x) + \kappa_y Y(y))}$$

where:

- $I(x, y)$ is the reconstructed image intensity.
- S_i is the i^{th} acquired k-space sample.
- O is the total number of k-space samples.
- DCF_i is the density compensation factor for sample i , determined by the sampling trajectory.
- $\kappa_{x,i}$ and $\kappa_{y,i}$ are the k-space coordinates of sample i .

However, in practice, the INUDFT is far too computationally inefficient [3]. So, this program, in addition to the INUDFT, used a method that MRI systems typically use called gridding to transform the k-space frequencies onto a uniform grid enabling a traditional IFFT, known as the Non-Uniform Fast Fourier Transform (NUFFT) [16] [17]. The algorithm reduces the number of computations from an estimated several billion with INUDFT to an estimated couple million with INUFFT [2] [16] [17].

The gridding transform works by expanding k-space by an

oversampling factor (OSF) and distributing even k-space sample to the surrounding cells proportionally to the distance from the sample to the new cell [2] [17].

The gridding transform is defined as:

$$G(m, n) = \sum_{i=1}^N DCF_i S_i K(m - g_{x,i}) K(n - g_{y,i})$$

Where:

- $G(m, n)$ is the Cartesian k-space grid
- S_i is the i^{th} non-uniform k-space sample
- DCF_i is the density compensation factor (DCF)
- $C(x)$ is the interpolation kernel
- $g_{x,i}$ and $g_{y,i}$ are the continuous grid coordinates of the sample after scaling to the oversampled grid
- N is the total number of k-space samples

The interpolation kernel that this program uses is the normalized Kaiser-Bessel kernel, defined as [16] [18] [19]:

$$K(\kappa) = \begin{cases} \frac{I_0\left(\beta \sqrt{1 - \left(\frac{2\kappa}{W}\right)^2}\right)}{I_0(\beta)}, & |\kappa| \leq \frac{W}{2}, \\ 0, & |\kappa| > \frac{W}{2} \end{cases}$$

Where:

- W is the kernel Width, usually 3-6
- $I_0(x)$ is the modified Bessel Function of the First Kind of Order Zero
- $\beta = \pi \sqrt{\left(\frac{W}{OSF}\right)^2 (OSF - \frac{1}{2})^2 - 0.8}$, Kaiser-Bessel Optimization found by Beatty et al. [18].
-

The image can then be reconstructed by taking the IFFT using the Bluestein Algorithm [13] and applying deapodization by multiplying the resultant image with the Inverse Fourier transform of the interpolation kernel (Kaiser-Bessel Function) [19]:

$$I(x, y) = F\{G(m, n)\} \times F^{-1}\{C(m, n)\}$$

Where:

- $I(x, y)$ is the reconstructed image
- $G(m, n)$ is the uniform k-space grid
- $C(m, n)$ is the interpolation kernel

Due to the NUFFT distributing the power of the non-uniform k-space samples to the surrounding cells, it is not an exact reconstruction like the NUDFT, it is at best an approximation [17].

Results were measured and compared using 2 standardized metrics. The first is Peak Signal-to-Noise Ratio (PSNR), used to compare how much the reconstruction matches the original image.

PSNR is defined as [9]:

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

$$MSE = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

Where:

- MAX_I is the maximum possible pixel value of the image
- m and n are the width and height of the image in pixels
- $I(i, j)$ is the original image
- $K(i, j)$ is the reconstructed image

The second is Structural Similarity Index Measure (SSIM), used to quantify structural similarity, which correlates better with perceived image quality than simple pixel-wise metrics [9]. This paper looks at the possible use of SSIM as a more appropriate quantitative metric for MR Imaging.

Since SSIM is a standard metric used in image processing, and for the sake of brevity, the formal definition of SSIM will be omitted. Interested readers should refer to [9] for further information.

Computational time was measured from the beginning of k-space reconstruction through completion of the reconstruction. Image preparation, trajectory generation, pre-calculations, and image saving or display were excluded from the reported reconstruction time. All final reconstructions were normalized to a common intensity range prior to calculation of PSNR and SSIM. Sampling percentages were defined relative to the total number of samples required for a fully sampled Cartesian reconstruction of the corresponding image.

For gridding-based NUFFT reconstruction, an oversampling factor (OSF) of 1.5 and a Kaiser–Bessel interpolation kernel width of 6.0 were used. The Kaiser–Bessel kernel parameter β was calculated using the formula described by Beatty et al. [18].

A fixed random seed of ‘12345’ was used for comparisons requiring identical sampling realizations, while separate seeds were used where variability between random sampling realizations was being evaluated.

All computations were performed on a computer equipped with an Intel i5-8400 processor, using Windows 10 with 32GB of RAM. Reconstruction algorithms were implemented in C++ with Visual Studio Code compiled using GCC through the MSYS2 UCRT64 environment. OpenCV was used for visualization and image processing [20].

IV. Results

A. NUFFT Performance Comparison

To determine whether the NUFFT provides a computational advantage over the original NUDFT implementation without significantly affecting reconstruction quality, both algorithms were evaluated using identical sampling trajectories and sampling percentages. Reconstruction time was measured to evaluate computational efficiency, while PSNR and SSIM were used to assess reconstruction quality. The results were then compared across increasing sampling percentages to determine how the performance difference changes with the number of samples acquired.

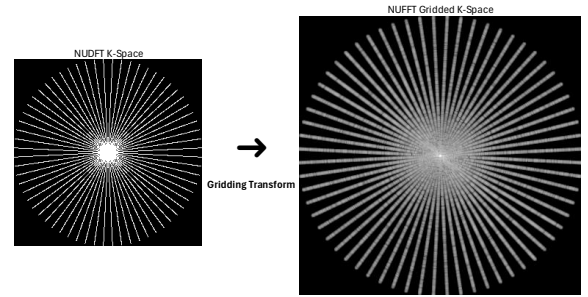


Figure 2. Effect of the gridding transform on k-space samples. Non-Cartesian k-space samples (left) are interpolated onto a Cartesian grid (right) prior to image reconstruction.

Sampling	NUDFT Time (ms)	NUFFT Time (ms)	PSNR Difference	SSIM Difference
10%	46409	23681	-2.9413	-0.118781
25%	106778	36563	0.5246	0.070377
50%	216182	60983	-0.2193	0.007504
75%	314174	82119	0.3138	0.00084
100%	418215	108273	-0.278	-0.005031

Figure 3. Comparison of average NUDFT and NUFFT reconstruction performance across sampling percentages. Reconstruction time, PSNR difference, and SSIM difference are reported for each sampling percentage. Difference measured as NUDFT – NUFFT.



Figure 4. Visual comparison of NUDFT and NUFFT reconstructions for the same k-space acquisition. Both methods use identical sampling conditions and reconstruction parameters.

The results show that the NUFFT algorithm provides a substantial computational efficiency improvement over the original NUDFT, achieving up to approximately 4× faster reconstruction times at higher sampling levels. Additionally, the NUFFT demonstrated minimal impact on reconstruction quality, with PSNR and SSIM showing little degradation and, in

some cases, improvement, particularly at sampling percentages above 10%. Overall, these results demonstrate that the NUFFT provides a favorable balance between computational efficiency and reconstruction quality. Therefore, the NUFFT will be used for the remaining trajectory, noise, and motion comparisons.

B. Trajectory Optimization

To ensure a fair comparison, it is important to determine the optimal parameter settings for each trajectory under a given reconstruction scenario. To accomplish this, a program was developed to systematically evaluate a range of trajectory parameters and measure the resulting reconstruction quality using peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM). The computational time required for each reconstruction was also recorded. The parameter combination producing the highest reconstruction quality was then selected for each trajectory and used in subsequent comparisons.

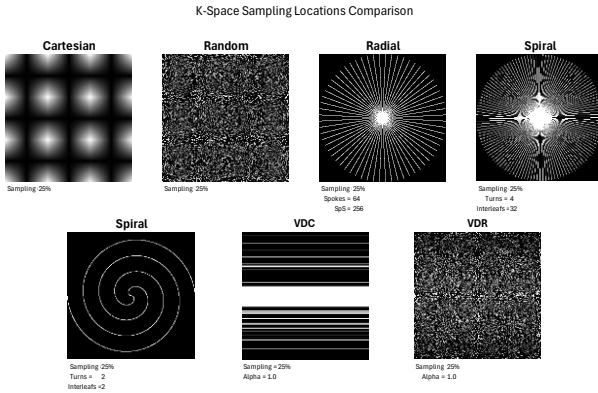


Figure 5. Sampling locations of the evaluated k-space trajectories. Each trajectory is shown at the same sampling percentage to illustrate differences in k-space coverage and sampling distribution. Note spiral trajectories often looks similar to radial trajectories when using high number of turns or interleaves, see the 2nd spiral k-space image.

1) Radial Optimization

The radial trajectory was first evaluated by varying the number of radial spokes $[0, R]$, and the number of samples per spoke. The results of this parameter sweep were used to identify the optimal configuration that produced the highest reconstruction quality.

256x256									
#Spokes	PSNR (dB)	SSIM	Heatmap	PSNR (dB)	SSIM	Heatmap	PSNR (dB)	SSIM	Heatmap
12	0.5090	0.1794	0.0190	0.0190	0.0190	0.0190	0.0190	0.0190	0.0190
256	0.7590	0.1895	0.1721	0.1895	0.1895	0.1895	0.1895	0.1895	0.1895
128	0.8421	0.4032	0.1145	0.4032	0.4032	0.4032	0.4032	0.4032	0.4032
64	0.8808	0.5177	0.1448	0.5177	0.5177	0.5177	0.5177	0.5177	0.5177
32	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259
16	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259
8	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259
4	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259
2	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259
1	0.8899	0.5259	0.1453	0.5259	0.5259	0.5259	0.5259	0.5259	0.5259

Figure 6. Radial trajectory reconstructions as a function of the number of spokes and number of samples per spoke. The left-side figures represent results

for a 256x256 image, and the right-side figures represent results for a 512x512 image. Higher PSNR values and SSIM values closer to 1.0 indicate better reconstruction quality. Computation time measured in milliseconds.

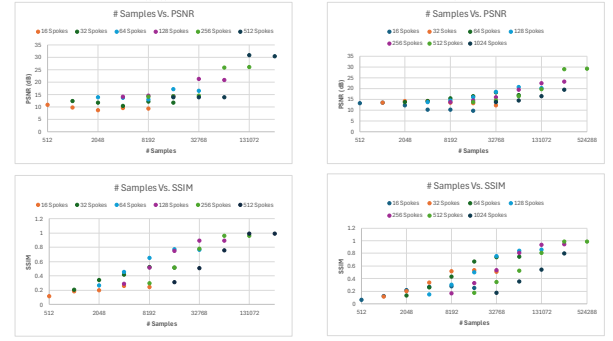


Figure 7. Radial trajectory reconstructions as a function of the total number of samples. Different colors indicate the distribution of the number of spokes. The left-side figures represent results for a 256x256 image, and the right-side figures represent results for a 512x512 image. Higher PSNR values and SSIM values closer to 1.0 indicate improved reconstruction quality.



Figure 8. Radial trajectory reconstructed images comparison. Shows the visual effects of number of samples per spoke. The top row images show results for a 256x256 image, while the bottom row images show results for a 512x512 image.

The results indicate that the optimal number of samples per spoke is equal to the resolution of one side. The number of spokes exhibits diminishing returns as the number of spokes increases, with the optimal number of spokes being approximately 25% of the resolution of one side. Therefore, 64 spokes with 256 samples per spoke were selected for comparisons at 25% sampling for a 256×256 image. Similarly, for 512×512 images sampling at 25%, 128 spokes with 512 samples per spoke were selected for further comparisons. For sampling levels greater than 25%, only the number of spokes was increased.

2) Spiral Optimization

The same optimization procedure was then applied to the spiral trajectory. The number of samples per interleaf and the number of turns were varied to determine their effects on reconstruction quality and computational time.

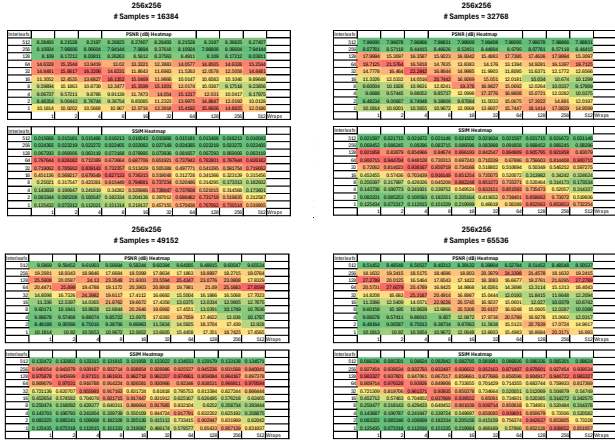


Figure 9. Spiral trajectory reconstructions for a 256x256 Lena test image, plotted as a function of the number of wraps and number of interleaves. Higher PSNR values and SSIM values closer to 1.0 indicate improved reconstruction quality.

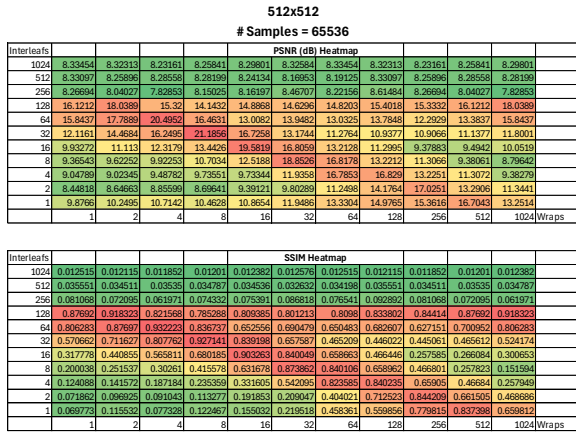


Figure 10. Spiral trajectory reconstructions for a 512x512 Lena test image, plotted as a function of the number of wraps and number of interleaves. Higher PSNR values and SSIM values closer to 1.0 indicate improved reconstruction quality.

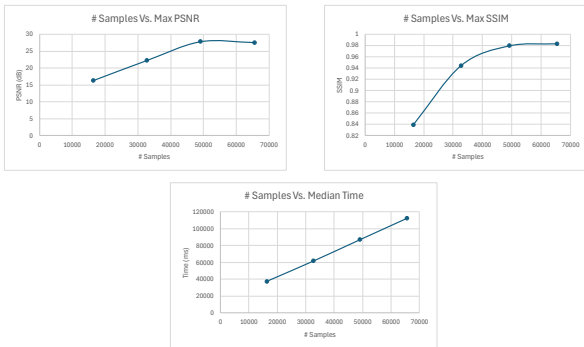


Figure 11. Spiral trajectory reconstructions as a function of the number of samples. Higher PSNR values and SSIM values closer to 1.0 indicate improved reconstruction quality. Computation time is measured in milliseconds.



Figure 12. Visual comparison of optimal spiral trajectory settings for each sampling scenario.

Optimal Spiral Settings					
# Samples	Optimal Interleaves	Optimal Wraps	PSNR (dB)	SSIM	Time (ms)
16,384	32	4	16.32	0.8391	37,360
32,768	32	4	22.28	0.9382	61,382
49,152	64	2	25.9	0.9703	87,184
65,536	64	2	27.61	0.9765	111,699

Figure 13. Table shows optimal spiral trajectory settings with their respective measurements. Note that, for values that were close, a visual comparison based on visual artifacts was made to inform the final decision.

The results show that the optimal spiral parameters vary with the total number of samples. However, the variation in PSNR and SSIM between sampling scenarios may be attributed to differences in the distribution of samples within k-space for the Lena test image. This is demonstrated by the decrease in both PSNR and SSIM when the total number of samples increased from 49,152 to 65,536. Although the number of samples increased, the resulting sampling pattern did not necessarily provide a more effective distribution of k-space information for this image. Across all four sampling scenarios, 32 interleaves with 4 wraps consistently produced one of the highest PSNR and SSIM values among the tested scenarios.

Therefore, 32 interleaves and 4 wraps were selected as the spiral trajectory configuration for further comparisons. Similarly, the data reflected that the optimal settings to use for a 512x512 image with 25% sampling were 32 interleaves with 8 wraps.

3) Variable Density Optimization

The optimization procedure was again performed on the Variable Density Cartesian trajectory. A sweep of 3 parameters was performed: acceleration factor, alpha value, and center percentage. Again, their effects on reconstruction quality were measured using PSNR and SSIM values, and the computation time was also recorded.

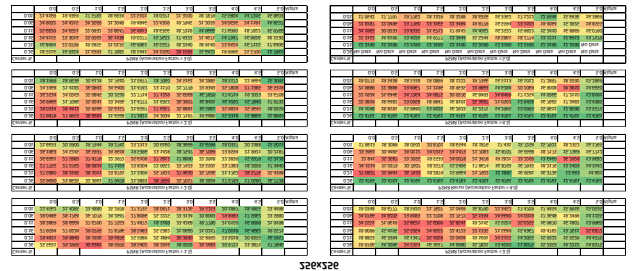


Figure 14. Cartesian Variable Density trajectory reconstruction PSNR as a function of center percentage and alpha value. Higher PSNR values indicate improved reconstruction quality.

256x256

Sample Percentage	Random	Cartesian	Radial	Spiral	VDR	VDC
10%	10.5142	8.61081	10.0301	11.524	9.9508	15.2177
20%	11.1525	11.6346	14.1155	16.0463	13.4803	23.3169
25%	14.1134	14.8477	17.2121	16.3206	11.526	39.5333
30%	14.1641	13.0028	17.3702	19.7156	13.0062	23.708
40%	18.5946	12.4722	13.3105	11.9417	30.5697	29.4448
50%	14.5215	9.90364	21.4814	22.2842	14.2688	36.7217
60%	14.0608	11.2128	26.5657	23.3629	13.6132	23.3657
70%	15.7992	10.9214	27.6233	24.4329	12.6331	26.3644
75%	14.9616	11.5405	24.2066	24.3962	12.9514	23.3021
80%	15.0906	12.5798	24.8444	24.8955	14.0235	26.3403
90%	18.919	16.5893	26.3937	26.2923	12.8778	32.3293
100%	14.8472	32.5558	25.8743	25.3187	14.8152	32.5558
110%	15.8962	31.3174	29.2432	29.5442	14.23	32.5558

Figure 15. Cartesian Variable Density trajectory reconstruction SSIM as a function of center percentage and alpha value. SSIM values closer to 1.0 indicate improved reconstruction quality.

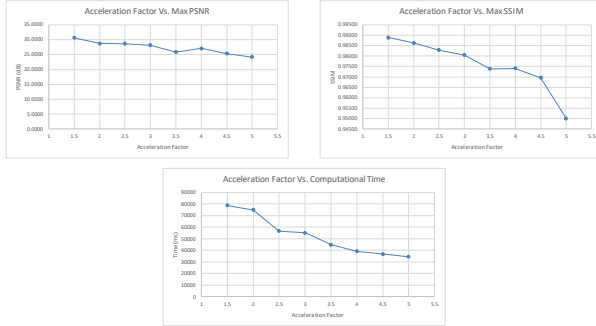


Figure 16. Cartesian Variable Density trajectory reconstruction metrics as a function of the number of samples. Higher PSNR values and SSIM values closer to 1.0 indicate improved reconstruction quality. Computation time is measured in milliseconds.



Figure 17. Cartesian Variable Density trajectory visual comparison of observed optimal configuration reconstruction.

Optimal Cartesian Variable Density Settings

Acceleration Factor	Optimal Center %	Optimal Alpha	PSNR (dB)	SSIM	Avg. Time (ms)
1.5	26%	1	30.64	0.9819	78.852
2	26%	2.5	28.77	0.9863	74.943
2.5	26%	1	28.67	0.9829	56.699
3	26%	3	28.16	0.9805	55.155
3.5	11%	1	25.86	0.9739	44.926
4	21%	1	26.74	0.9734	39.110
4.5	16%	2.5	25.37	0.9697	25.753
5	11%	1	23.83	0.9443	34.619

Figure 18. Table shows optimal Cartesian Variable Density trajectory settings with their respective measurements. Note, alpha values of 0.0 were not considered because it effectively disables variable density resulting in more volatile PSNR and SSIM values based on the random seed.

The results showed for acceleration factors < 2.5, the optimal center percentage was between 16% and 26%, with the optimal alpha value being between 0.0 and 1.0. For acceleration factors > 2.5 the optimal center percentage was between 6% and 16%, with the optimal alpha value being between 0.5 and 1.5.

The values measured may have varied between acceleration value runs due to the random seed used during generation. This may have caused some runs to produce higher or lower PSNR or SSIM values because a more favorable seed was used.

Based on consistency across each configuration scenario, an alpha value of 1.0 and a center percentage of 16% were selected for further Variable Density comparisons for all acceleration factors. The same configuration was also selected for 512x512 images and Random Variable Density trajectories.

C. Trajectory Comparison

Following the trajectory optimization, the optimized parameters were used to establish a consistent basis for comparing the different sampling trajectories. The trajectories were evaluated under the same reconstruction conditions and across varying sampling percentages to assess differences in reconstruction quality, sampling efficiency, and computational performance. In addition, noise and simulated motion were introduced to evaluate the robustness of each trajectory under conditions that may occur during practical MRI acquisition.

256x256

Sample Percentage	Random	Cartesian	Radial	Spiral	VDR	VDC
10%	10.5142	8.61081	10.0301	11.524	9.9508	15.2177
20%	11.1525	11.6346	14.1155	16.0463	13.4803	23.3169
25%	14.1134	14.8477	17.2121	16.3206	11.526	39.5333
30%	14.1641	13.0028	17.3702	19.7156	13.0062	23.708
40%	18.5946	12.4722	13.3105	11.9417	30.5697	29.4448
50%	14.5215	9.90364	21.4814	22.2842	14.2688	36.7217
60%	14.0608	11.2128	26.5657	23.3629	13.6132	23.3657
70%	15.7992	10.9214	27.6233	24.4329	12.6331	26.3644
75%	14.9616	11.5405	24.2066	24.3962	12.9514	23.3021
80%	15.0906	12.5798	24.8444	24.8955	14.0235	26.3403
90%	18.919	16.5893	26.3937	26.2923	12.8778	32.3293
100%	14.8472	32.5558	25.8743	25.3187	14.8152	32.5558
110%	15.8962	31.3174	29.2432	29.5442	14.23	32.5558

Figure 19. Standard test image Lena (256x256) trajectory PSNR (left), measured in dB, and SSIM (right) comparisons. Displayed as a function of number of samples represented as a percentage of the total number of samples required to fully reconstruct the image. Higher PSNR values indicate better reconstruction. SSIM values closer to 1.0 indicate better structural reconstruction.

256x256

Computation Time

Sample Percentage	Random	Cartesian	Radial	Spiral	VDR	VDC
10%	21491	24434	21945	22327	22347	22462
20%	33784	33503	32114	31734	33067	33903
25%	37824	36810	37558	36979	38099	38633
30%	41685	43848	43149	41573	41864	41902
40%	53023	53691	52001	53389	64072	51816
50%	62831	64143	63973	66504	77140	67298
60%	72736	73256	73022	71706	89988	72080
70%	82172	83251	81539	86718	102548	85262
75%	86284	88142	88585	89894	109596	90438
80%	92255	93593	92754	91864	115952	97063
90%	101024	102242	102292	100819	130020	104052
100%	118229	114738	113344	114589	141863	114663
110%	121091	122488	119719	128278	155189	113119

Figure 20. Standard test image Lena (256x256) trajectory computation time comparison. Time measured in milliseconds (ms).

Random and Variable Density Random (VDR) both produced poor reconstructions, with VDR yielding better results but still poor in comparison to the other trajectories. Regular Cartesian presented severe ghosting in the reconstructed images with low PSNR and SSIM values for all undersampled runs.

While Cartesian Variable Density (VDC) generally produced the highest PSNR and SSIM values, a visual comparison of

lower sampling percentages shows that VDC yielded blurrier images with higher PSNR and SSIM, whereas radial and spiral trajectories produced images with lower metrics but with much greater clarity.

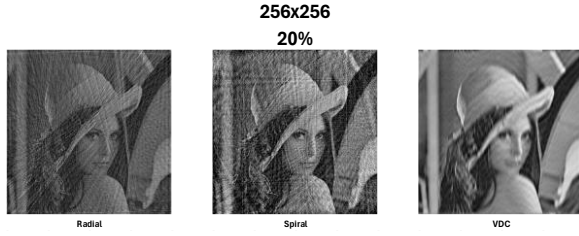


Figure 21. Comparison of radial, spiral, and Cartesian Variable Density (VDC) sampled at 20%. Radial and spiral had lower PSNR and SSIM values but better clarity, VDC has higher values but less clarity.

This demonstrates a limitation of using PSNR and SSIM as the sole measures of reconstruction quality in this study. While PSNR and SSIM provide a general approximation of MRI image trajectory quality, a visual comparison is necessary for a final decision.

Considering the above, VDC generally performed the best of all sample percentages. However, radial and spiral both performed comparably, particularly for lower sampling percentages ($<50\%$), while VDC produced higher PSNR and SSIM values the images were blurry and would be difficult to read for a real MRI image. Radial and Spiral produced much clearer images.

For sampling percentages $>50\%$, VDC seemed to perform only marginally better compared to spiral and radial. VDC yielded higher PSNR and SSIM than radial and spiral, however ghosting started to appear at sampling percentages below 75%, whereas radial and spiral produced images only with slight visual artifacts. And again, radial and spiral produced images with greater clarity, although the difference in clarity was smaller at sampling percentages below 50%. For fully sampled images, regular Cartesian and VDC performed the same, as to be expected.



Figure 22. Comparison of spiral, Cartesian Variable Density, and Cartesian trajectories sampled at 50%. Images are an example of ghosting and aliasing.

Computation time was consistent across most of the trajectories, with the exception of the Random Variable Density (VDR) which produced slightly longer computation times. This is likely due to inefficiencies in the VDR algorithm, as each sample had to be generated and tested for probability individually. The trajectory was mainly implemented as a control as a purely random trajectory would not be practical in a

real MRI system.

Radial and spiral trajectories both performed very well and similarly. However, spirals generally produced reconstructions with less intense visual artifacts, and better clarity than radial.

Based on this experiment, spiral can be considered the most effective trajectory for sampling percentages below 50% as the best trajectory out of the six. For sampling percentages $>50\%$, VDC and Spiral performed very similarly, and a confident conclusion cannot be drawn, as MRI systems deal with many different variables than just reconstruction quality, such as motion and noise.

1) Multi-Image Comparison

The above experiment was only for the Lena standard test image; to draw a more definite conclusion, trajectories were compared across multiple images and their PSNR and SSIM values were averaged to reduce the influence of trajectory performance on any particular image.

Reconstruction time was recorded for each experiment; however, direct comparison of computation times across images was not performed because the test images had different resolutions and therefore different numbers of k-space samples at the same sampling percentage.

Average PSNR (dB)						
Sampling	Cartesian	Random	Radial	Spiral	VDC	VDR
25%	11.9 ± 2.73	11.05 ± 1.95	17.17 ± 1.81	18.32 ± 3.59	21.4 ± 6.55	11.13 ± 1.26
50%	11.57 ± 1.44	12.58 ± 1.74	20.12 ± 1.29	21.17 ± 4.41	24.78 ± 7.03	13.07 ± 3.03
75%	13.45 ± 2.24	12.89 ± 1.7	21.77 ± 2.25	22.43 ± 5.36	29.22 ± 6.73	12.52 ± 2.29
100%	25.52 ± 6.35	13.79 ± 1.83	22.9 ± 3.05	23.23 ± 6.19	32.4 ± 6.98	13.07 ± 2.62

Average SSIM						
Sampling	Cartesian	Random	Radial	Spiral	VDC	VDR
25%	0.221 ± 0.263	0.146 ± 0.202	0.765 ± 0.127	0.837 ± 0.111	0.849 ± 0.22	0.297 ± 0.148
50%	0.372 ± 0.137	0.343 ± 0.215	0.88 ± 0.063	0.895 ± 0.087	0.919 ± 0.103	0.407 ± 0.162
75%	0.603 ± 0.133	0.437 ± 0.152	0.912 ± 0.058	0.908 ± 0.084	0.968 ± 0.047	0.41 ± 0.232
100%	0.948 ± 0.049	0.541 ± 0.182	0.936 ± 0.046	0.912 ± 0.085	0.977 ± 0.049	0.389 ± 0.231

Figure 23. The table shows the average PSNR and SSIM values from seven different test images (see next figure), with the calculated standard deviation.

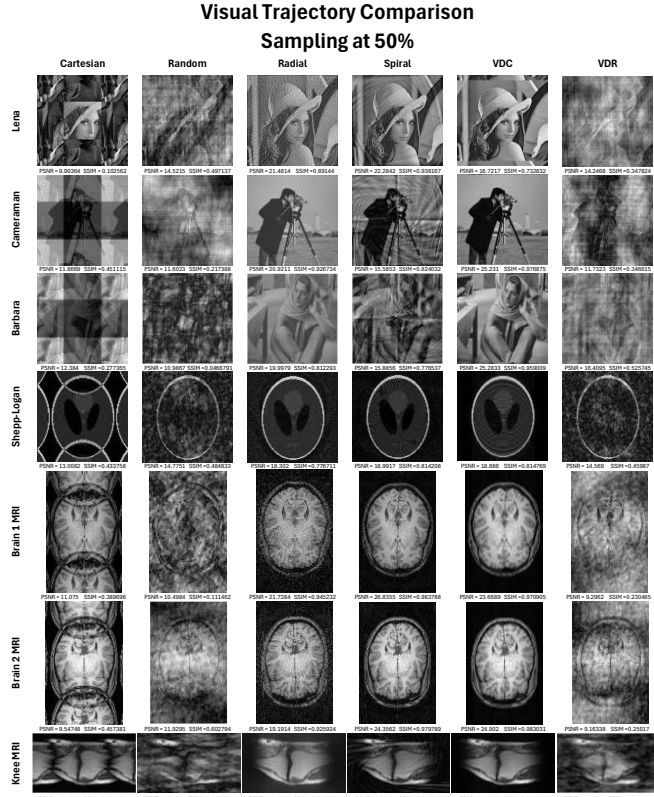


Figure 24. Visual comparison of reconstructions from each trajectory. The test images included Lena, Barbara, Cameraman, the Shepp-Logan MRI test, 2 real brain MRIs obtained from OpenNeuro [21], and 1 real knee MRI reconstructed from k-space acquired from NYU FastMRI dataset [22].

The average values recorded in Figure 25 show Cartesian Variable Density outperformed the other trajectories particularly at higher sampling percentages. Again, radial and spiral performed similarly, yet both yielded lower average PSNR and SSIM values compared to VDC. However, VDC yielded the highest standard deviation across all trajectories.

However, a visual comparison once again shows VDC is best at preserving overall reconstruction, while radial and spiral preserve fine detail better than VDC. It is particularly evident in the Shepp-Logan MRI test image’s small ellipses and the area near the posterior horns of the lateral ventricles in the brain MRI images.

Unexpectedly, regular random trajectory outperformed Random Variable Density. This is probably again due to the pure random nature of both trajectories causing the variable density to have minimal effect on a purely random trajectory.

Cartesian trajectory performed almost as expected, reconstructions showed severe ghosting caused by the evenly spaced skipped k-space lines. Note, for the fully sampled reconstructions Cartesian unexpectedly underperformed on some images.

Motion and Noise Resistance

The above tests assume ideal conditions; the patient does not move and assumes the machine operates in a perfectly controlled environment. However, real MRI systems have to deal with patients breathing, twitching, and other small movements [23], as well as performing in noisy conditions [24].

The k-space trajectory used during acquisition could have a significant impact on how much movement and noise affects the overall reconstruction quality [23]. To simulate movement, a spatial shift was introduced partway through the k-space acquisition. A separate test on noise robustness will also be done by adding a simple complex Gaussian noise to the k-space samples.

2) Motion Resistance

To implement spatial shifts into the program, we take advantage of the shift property of the Fourier transform, which states that a spatial translation corresponds to a linear phase shift in the spectral domain [10]:

$$F\{f(x - x_0)\} = e^{-ikx_0}F(k)$$

Random and Variable Density Random were previously shown to be far inferior to the other trajectories and were therefore excluded from the following motion and noise comparisons. Additionally, to ensure a fair comparison, the following comparisons keep the random seed of Cartesian Variable Density constant throughout all runs.

Sampling 50%, Shift by 0.05

PSNR (dB)

Onset	Cartesian	Radial	Spiral	VDC
25%	9.502	15.311	13.952	12.648
50%	9.978	17.613	17.702	15.570
75%	9.940	19.791	17.609	26.185
Baseline	9.904	21.481	22.284	27.267

SSIM

Onset	Cartesian	Radial	Spiral	VDC
25%	-0.0148	0.5135	0.4510	0.2764
50%	0.0513	0.7084	0.7383	0.5973
75%	0.1014	0.8296	0.7977	0.9752
Baseline	0.1026	0.8914	0.9382	0.9804

Figure 26. PSNR and SSIM values from baseline reconstruction with the same settings without movement. Onset time is a percentage of total acquisition time that represents at what time the shift occurs.

Sampling 50%, Shift by 0.05

 Δ PSNR (dB)

Onset	Cartesian	Radial	Spiral	VDC
25%	-0.402	-6.171	-8.332	-14.620
50%	0.074	-3.868	-4.582	-11.697
75%	0.036	-1.690	-4.675	-1.082

 Δ SSIM

Onset	Cartesian	Radial	Spiral	VDC
25%	-0.117	-0.378	-0.487	-0.704
50%	-0.051	-0.183	-0.200	-0.383
75%	-0.001	-0.062	-0.141	-0.005

Figure 27. Change in PSNR and SSIM values from a control image reconstructed with the same settings without movement. Onset time is a percentage of total acquisition time that represents at what time the shift occurs.

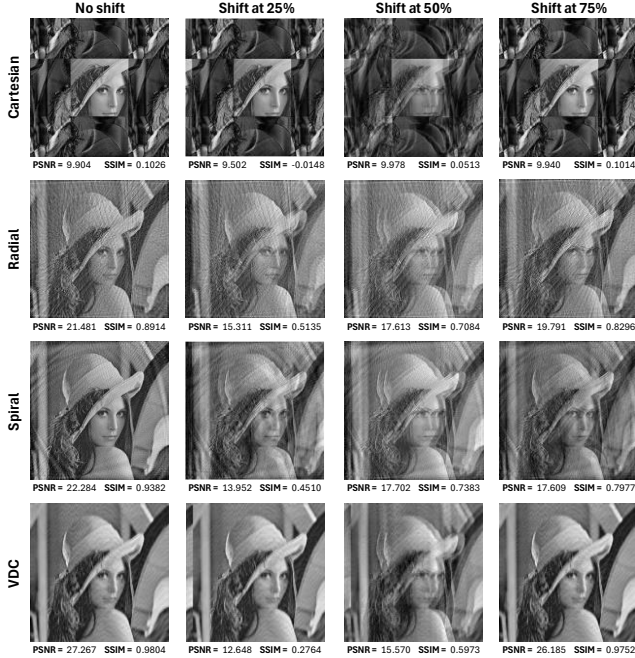


Figure 28. Visual comparison between trajectories shows image degradation as onset time changes.

The results show that the Cartesian trajectory produced the smallest change in PSNR and SSIM values when motion was introduced. However, the original Cartesian reconstruction was already relatively poor due to undersampling. The additional blurring and ghosting caused by the simulated motion may have spread some of the more accurate pixels across the image, resulting in a small increase in PSNR and SSIM rather than indicating improved motion resistance. This was further

investigated using the fully sampled comparison shown in Figure 29.

Sampling 100%, Shift by 0.05

 Δ PSNR (dB)

Onset	Cartesian	Radial	Spiral	VDC
25%	-20.085	-10.578	-11.242	-20.085
50%	-16.370	-8.234	-7.517	-16.370
75%	-3.047	-4.779	-6.329	-3.047

 Δ SSIM

Onset	Cartesian	Radial	Spiral	VDC
25%	-0.7202	-0.4165	-0.5027	-0.7202
50%	-0.3432	-0.2103	-0.2115	-0.3432
75%	-0.0040	-0.0728	-0.1322	-0.0040

Figure 30. Change in PSNR and SSIM values from a control image reconstructed with the same settings without movement. Onset time is a percentage of total acquisition time that represents at what time the shift occurs.

Considering the above, the radial trajectory generally produced the smallest change in PSNR and SSIM for motion occurring earlier in the acquisition. This suggests that radial sampling was relatively resistant to the simulated motion under the conditions tested.

Spiral sampling also showed resistance to motion, although generally not to the same extent as radial at lower sampling levels. Although the radial and spiral trajectories can appear visually similar, they acquire k-space differently. The spiral trajectory begins near the center of k-space and moves toward higher spatial frequencies, meaning that a motion event occurring later in the acquisition can primarily affect the higher frequency information. In contrast, each radial spoke contains both low and high frequency information, which may make the reconstruction less dependent on the timing of the motion.

Cartesian Variable Density (VDC) produced the smallest change in PSNR and SSIM when the simulated motion occurred at 75% of the total acquisition time. However, at 25% and 50% of the acquisition time, VDC produced some of the largest changes. One possible explanation is that the trajectory had not yet acquired enough of the important central k-space when the motion occurred early in the acquisition. Sampling the central frequencies earlier could potentially reduce this effect, although doing so could also change the acquisition efficiency and overall sampling distribution.

3) Noise Resistance

To evaluate trajectory robustness under simulated noisy acquisition conditions, Gaussian noise was introduced into the simulated k-space data prior to reconstruction. The resulting images were then compared with their corresponding noise-free reconstructions using PSNR and SSIM. This analysis was used to determine how effectively each sampling trajectory maintained reconstruction quality as the noise level increased.

The signal-to-noise ratio (SNR) convention used is defined as:

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

SNR was expressed in decibels as the ratio of signal power to noise power, with 0 dB representing equal signal and noise power. Complex Gaussian noise was added independently to the real and imaginary components of each k-space sample [25] [26].

Simulated Noise Addition Process

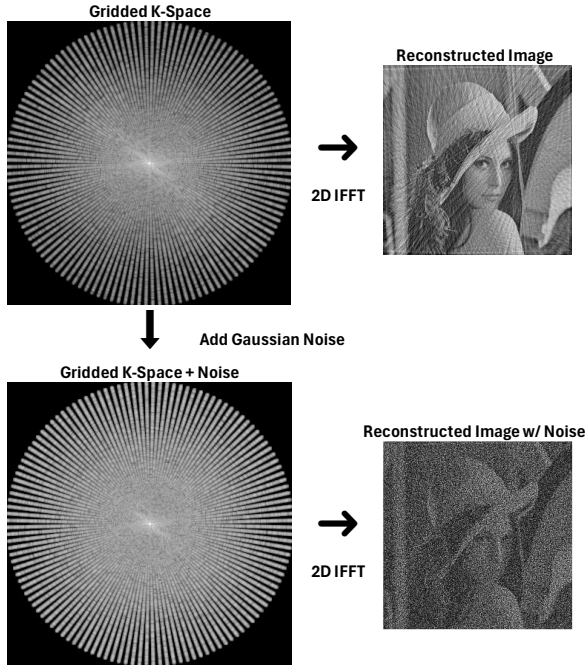


Figure 31. Simulated k-space noise addition and reconstruction process. Clean k-space data are combined with noise at a specified SNR to produce noisy k-space data for reconstruction.

Sampling 50%, Shift by 0.05

PSNR (dB)

SNR	Cartesian	Radial	Spiral	VDC
0.00	9.63275	9.879	9.52978	10.6986
10.00	9.83947	9.74135	9.91976	14.8878
20.00	9.95765	9.7961	10.1612	24.0219
30.00	9.92535	11.7534	11.6757	26.8333
40.00	9.90468	16.7354	17.2864	26.9296
None	9.90364	21.4814	22.2842	27.2674

SSIM

SNR	Cartesian	Radial	Spiral	VDC
0.00	0.044068	0.019633	0.016156	0.200698
10.00	0.096646	0.023231	0.021705	0.681638
20.00	0.103696	0.063556	0.074198	0.945457
30.00	0.103541	0.34192	0.370586	0.977914
40.00	0.102616	0.750335	0.815741	0.97989
None	0.102562	0.89144	0.938167	0.980352

Figure 32. Table shows PSNR and SSIM values of reconstruction at varying levels of noise. SNR is the Signal to Noise Ratio measure in decibels.

Sampling 50%

Δ PSNR (dB)

SNR (dB)	Cartesian	Radial	Spiral	VDC
0.00	-0.27089	-11.6024	-12.7544	-16.5688
10.00	-0.06417	-11.7401	-12.3644	-12.3796
20.00	0.05401	-11.6853	-12.123	-3.2455
30.00	0.02171	-9.728	-10.6085	-0.4341
40.00	0.00104	-4.746	-4.9978	-0.3378

Δ SSIM

SNR (dB)	Cartesian	Radial	Spiral	VDC
0.00	-0.05849	-0.87181	-0.92201	-0.77965
10.00	-0.00592	-0.86821	-0.91646	-0.29871
20.00	0.001134	-0.82788	-0.86397	-0.0349
30.00	0.000979	-0.54952	-0.56758	-0.00244
40.00	5.4E-05	-0.14111	-0.12243	-0.00046

Figure 33. Shows change in PSNR and SSIM values from a control image reconstructed with the same settings without noise. SNR is the Signal to Noise Ratio measure in decibels.

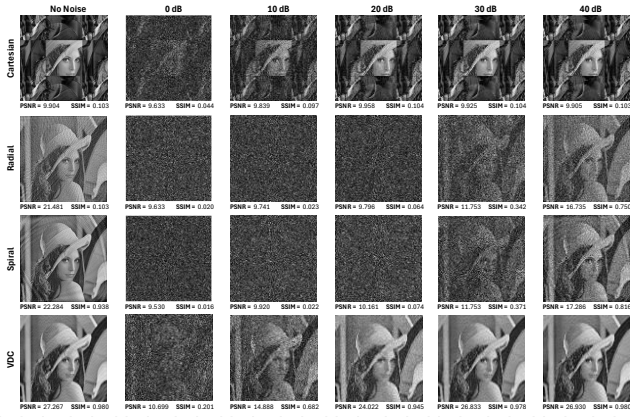


Figure 34. Visual comparison of trajectories that shows image degradation at different SNR levels.

The results show that Cartesian and VDC provide the greatest resistance to noise, while radial and spiral show severe image degradation across all SNR values, particularly at SNR values below 30dB. The radial and spiral trajectory noise image degradation is especially apparent when looking at the Δ SSIM values and looking at the visual comparison.

4) Noise and Movement Comparison

Generally, MRI systems operate in the presence of motion and noise. In the prior comparisons, radial and spiral trajectories produced greater motion robustness than the other trajectories but showed greater susceptibility to noise. Therefore, to test how each trajectory performs while dealing with motion and noise simultaneously, both previous motion and noise simulation methods were combined with varying levels of SNR values and onset times.

Movement + Noise Comparison

SNR = 40dB, Onset at 35%

Metric	Cartesian	Radial	Spiral	VDC
Δ PSNR	-0.3482	-7.1142	-9.0496	-14.6239
Δ SSIM	-0.11651	-0.44761	-0.47899	-0.69819

SNR = 30dB, Onset at 35%

Metric	Cartesian	Radial	Spiral	VDC
Δ PSNR	-0.31354	-10.3476	-11.1925	-14.459
Δ SSIM	-0.1165	-0.69961	-0.72209	-0.69894

SNR = 40dB, Onset at 65%

Metric	Cartesian	Radial	Spiral	VDC
Δ PSNR	-0.11323	-5.0699	-7.4011	-1.5875
Δ SSIM	-0.00578	-0.21335	-0.30127	-0.01189

Figure 35. Shows change in PSNR and SSIM values in different sampling scenarios. SNR level and shift onset time were varied. Onset time is a percentage of total acquisition time, which represents where the shift occurs.

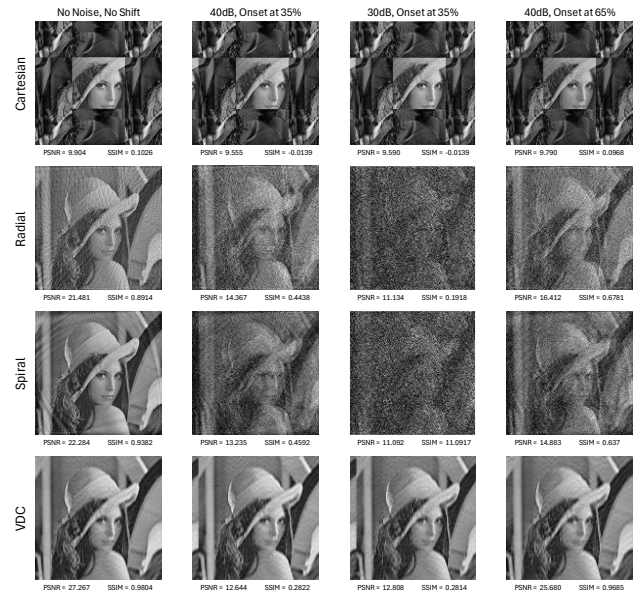


Figure 36. Visual comparison of trajectories shows image degradation at varying the SNR levels and onset times of the shift.

Across all the scenarios tested Cartesian and VDC showed the least image degradation. The previous experiments show that most of the image degradation comes from the noise.

V. Discussion

A. Radial Optimization

As seen in Figure 37, a visual comparison of the radial optimizations shows that when using a large number of spokes, the number of samples per spoke must be at least one-half of the resolution of one side to avoid severe artifacts. One possible explanation is that the spectral energy of the test image is not distributed uniformly throughout k-space. Consequently, when a radial spoke passes through a region with relatively high spectral power, those samples may contribute disproportionately to the reconstruction and amplify the resulting artifacts [27]. An offset of $\frac{\Delta\theta}{2}$ was added to help reduce this effect; however, as the number of spokes increases, the effect of the offset becomes negligible.

Additionally, this study used one convention for radial sampling, where each spoke was sampled from $[0, R]$. Another commonly used convention samples from $[-R, R]$, which effectively reduces the number of required spokes by half. This convention has its own advantages and disadvantages, but it is worth noting to avoid confusion when comparing radial trajectories. Another possible improvement would be the use of golden-angle sampling. Golden-angle radial trajectories use angular increments derived from the golden ratio and have been shown to provide benefits for radial MRI acquisition [28]. The specific effects and potential benefits of golden-angle sampling were outside the scope of this study and are left for future research.

B. Spiral Optimization

For the spiral optimization, 32 interleaves and 4 wraps were selected for further comparisons. However, in practice, using a large number of interleaves may reduce one of the primary advantages of spiral trajectories: their ability to perform single-shot acquisitions. Multi-shot trajectories may require additional time between acquisitions to return to their starting position or otherwise reset the trajectory. In contrast, a single-interleaf spiral can continuously traverse k-space, potentially reducing acquisition time and limiting the effects of motion during the acquisition [29]. Therefore, it may be beneficial in practice to reduce the number of interleaves and increase the number of wraps. However, as observed in the results, increasing the number of interleaves can improve reconstruction quality, suggesting that some number of interleaves may provide a useful tradeoff.

Based on the results of this study, configurations using fewer interleaves, potentially between 1 and 16, with a greater number of wraps, potentially between 64 and 512, could be investigated as a starting point for further optimization. These ranges should not be considered definitive operating parameters, since acquisition time, gradient constraints, motion sensitivity, and other hardware limitations were not directly evaluated in this study. A more complete optimization would need to consider these physical constraints in addition to reconstruction quality.

C. Variable Density Optimization

The variable-density Cartesian optimization demonstrated that the optimal sampling parameters depended on the acceleration factor. For acceleration factors < 2.5 , the optimal center percentage was generally between 16% and 26%, with optimal α values between approximately 0.0 and 1.0. At acceleration factors > 2.5 , the optimal center percentage decreased to approximately 6%–16%, while the optimal α increased to approximately 0.5–1.5. This suggests that the sampling distribution becomes increasingly important as the available number of samples decreases.

The decrease in optimal center percentage at higher acceleration factors can be explained by the tradeoff between sampling the central and peripheral regions of k-space. The center of k-space contains mostly low-spatial-frequency information and therefore has a strong influence on overall image contrast and structure. However, allocating a large percentage of the available samples to the center leaves fewer samples for higher frequencies, which contain information important for image detail and sharpness. At higher acceleration factors, where the total number of acquired samples is already limited, dedicating a large fraction of these samples to the center may therefore result in insufficient higher frequency k-space coverage. The observed decrease in center percentage with increasing acceleration is consistent with the need to balance these competing requirements.

The results also suggest an interaction between center percentage and α . The center percentage determines the size of the fully sampled central region, while α controls the degree to which the remaining samples are concentrated toward the

center. Increasing α can therefore provide additional emphasis on central k-space even when the fully sampled region is relatively small. This may explain why higher α values generally performed better at higher acceleration factors, where maintaining adequate sampling of the central region becomes more important while the total sampling budget is reduced. However, because the optimal values varied between configurations, these trends should be interpreted as general relationships rather than strict parameter requirements.

Variation between individual acceleration-factor runs was also observed. Because the variable-density sampling pattern contains a random component, different random seeds can produce different distributions of samples. Consequently, two configurations with identical acceleration, α , and center percentage can produce somewhat different PSNR and SSIM values, depending on the bias of that particular seed. This variation means that the configuration producing the single highest metric for a particular run may not necessarily represent the most robust choice across different sampling realizations. Additionally, $\alpha = 0$ was excluded from the final optimization because it effectively removes the variable-density component of the trajectory, making the resulting sampling pattern more dependent on the random seed.

Although different acceleration factors showed different individual optima, an α value of 1.0 and a center percentage of 16% were selected for subsequent comparisons. These parameters provided consistently strong reconstruction quality across the tested acceleration factors while avoiding the need to optimize the trajectory independently for every condition. The selected values should therefore be used as a base configuration for the future that could be improved rather than as universally optimal VDC parameters. The same configuration was also applied to the 512×512 images and the random variable-density Cartesian trajectory to evaluate whether the observed performance was maintained under different test conditions.

D. Trajectory Comparison

In the multi-image comparison, VDC yielded the highest average PSNR and SSIM values across the trajectories, particularly at higher sampling percentages. However, VDC also produced the highest standard deviation. This may be due to the random nature of VDC, where different random seeds can produce more or less favorable sampling patterns for a particular image.

A visual comparison once again shows that VDC generally does the best at preserving the overall image, while radial and spiral trajectories preserve fine details better than VDC. This is particularly evident in the small ellipses of the Shepp-Logan MRI test image and in the area near the posterior horns of the lateral ventricles in the brain MRI images.

Unexpectedly, the regular random trajectory outperformed the Random Variable Density trajectory. One possible explanation is that the random sampling already produces a relatively dispersed distribution of samples, limiting the

additional benefit of applying a variable-density distribution to an already random trajectory.

The Cartesian trajectory performed mostly as expected, with undersampled reconstructions showing severe ghosting caused by the evenly spaced skipped k-space lines. However, the fully sampled Cartesian reconstructions unexpectedly underperformed on some images. Further investigation indicated that this was related to how the trajectory handles non-square images and possible rounding errors when the image dimensions do not produce an exact square sampling grid. Despite these implementation limitations, the disadvantages of Cartesian undersampling were clear, with reduced PSNR and SSIM and severe ghosting in the undersampled reconstructions.

E. Motion Comparison

In the motion resistance comparison, radial appeared to show the greatest resistance to the simulated motion among the non-Cartesian trajectories. This may be related to how radial sampling acquires one spoke at a time, with each spoke containing both low and high frequency information. As a result, a motion event affecting one portion of the acquisition may be partially balanced by information acquired from other spokes. Radial trajectories also contain angular redundancy, where nearby spokes sample similar regions of k-space. Therefore, if one spoke is affected by motion, nearby spokes may help reduce the overall effect of the corrupted data [28]. Increasing the number of spokes could potentially increase this redundancy, although this was not directly investigated in this study.

The spiral trajectory also showed significant resistance to the simulated motion. One possible explanation is the order in which spiral trajectories acquire k-space. The trajectory generally begins near the center and moves toward higher spatial frequencies. Therefore, a motion event occurring later in the acquisition may primarily affect higher-frequency information. In contrast, radial sampling acquires both low- and high-frequency information within each spoke, which may make it less dependent on when the motion occurs.

However, at higher sampling levels, spiral appeared to become more resistant to motion than radial. One possible explanation is that increasing the number of interleaves changes the distribution of high-frequency samples throughout the acquisition. Increasing the number of interleaves may therefore provide additional opportunities to acquire important frequencies before a motion event occurs. However, increasing the number of interleaves could also increase acquisition time, so this would need to be evaluated alongside realistic acquisition constraints [28] [30].

VDC showed relatively high sensitivity to motion when the shift occurred during the first half of the acquisition, but relatively low sensitivity when the shift occurred during the latter portion. One possible explanation is that the VDC trajectory used in this study sampled the k-space lines

sequentially from the top of the grid, meaning that the central frequencies were not acquired until later in the acquisition. A motion event occurring early may therefore affect the reconstruction before the most important central frequencies have been acquired. One potential solution would be to acquire the central frequencies earlier in the acquisition. However, changing the acquisition order could introduce additional acquisition constraints and would need to be evaluated to determine whether the potential improvement in motion resistance justifies the change.

F. Noise Comparison

In the noise resistance test, Cartesian and VDC trajectories demonstrated the greatest resistance to noise, while radial and spiral trajectories showed greater degradation in image quality as noise increased. This result was unexpected because radial and spiral trajectories contain multiple closely spaced samples along different spokes or turns of the trajectory. It was initially hypothesized that this sampling redundancy could provide an averaging effect and reduce the influence of noise during reconstruction. However, redundant sampling does not necessarily reduce noise unless the reconstruction process effectively combines the redundant information.

One possible explanation for the observed behavior is the difference between Cartesian and non-Cartesian reconstruction. Cartesian and VDC trajectories sample directly on the Cartesian k-space grid, allowing each acquired sample to correspond to a defined grid location without requiring the same degree of interpolation used for radial and spiral trajectories. In contrast, radial and spiral trajectories require the non-Cartesian samples to be interpolated or gridded onto a Cartesian grid during reconstruction. This additional processing can redistribute noise and introduce spatially correlated noise, particularly when the sampling density is nonuniform. Therefore, the higher noise sensitivity observed for radial and spiral trajectories may be related not only to the number of samples, but also to how those samples are distributed and subsequently mapped onto the reconstruction grid.

Another possible explanation for the increased noise sensitivity of radial and spiral trajectories is the use of a density compensation function (DCF). The DCF compensates for the nonuniform sampling density of these trajectories. In this implementation, the compensation weights are related to the local sampling density and are derived from the Jacobian of the sampling trajectory. Consequently, samples in regions with lower sampling density can receive greater weight during reconstruction. While this weighting can improve the accuracy of the noiseless reconstruction, it may also increase the contribution of noise from those samples.

This possibility was investigated by comparing radial and spiral reconstructions with and without DCF weighting. Figure 35 shows the PSNR and SSIM of noisy reconstructions relative to their corresponding noiseless reconstructions. The comparison was made against the noiseless reconstruction rather than the original test image so that the measured change

in PSNR and SSIM primarily reflects the degradation caused by noise rather than differences in the underlying reconstruction quality.

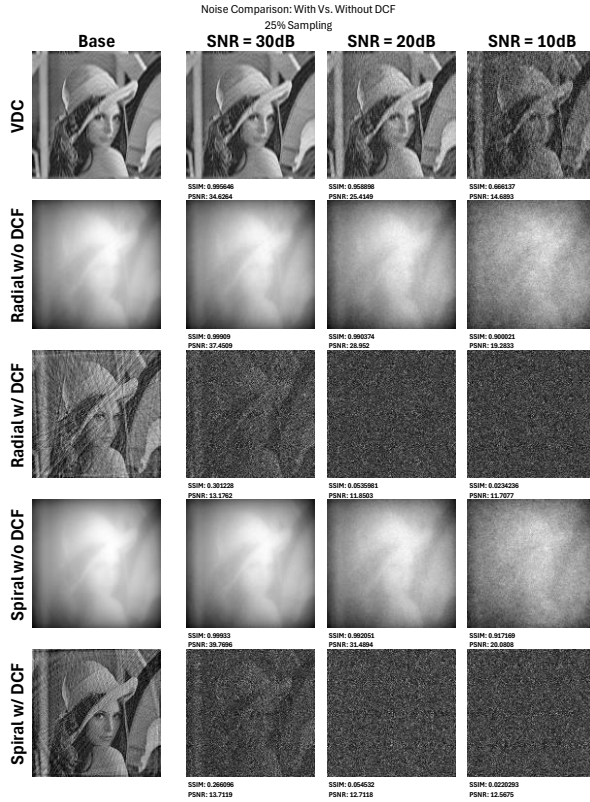


Figure 38. Comparison of image degradation with and without DCF weighting for radial and spiral trajectories. PSNR and SSIM are measured relative to the corresponding noiseless reconstruction at 25% sampling.

As shown in Figure 39, applying the DCF significantly improves the base reconstruction for both radial and spiral trajectories. However, these same reconstructions exhibited a greater decrease in PSNR and SSIM as noise was introduced. Without DCF weighting, radial and spiral trajectories produced lower-quality baseline reconstructions, but their PSNR and SSIM decreased less as the noise level increased. This suggests that the DCF improves reconstruction fidelity by correcting for nonuniform sampling density, but this correction may also increase the sensitivity of the reconstruction to acquisition noise.

Additionally, as shown in Figure 40, radial and spiral trajectories without DCF weighting showed greater resistance to noise than the VDC trajectory. This provides some evidence that the sampling redundancy present in radial and spiral trajectories may contribute to noise resistance. However, because the trajectories also differ in their k-space sampling distributions and reconstruction methods, this result does not isolate sampling redundancy as the sole cause of the increased noise resistance.

Overall, these results indicate that sampling redundancy alone is not sufficient to predict noise resistance. Instead, noise sensitivity appears to depend on the interaction between the

sampling trajectory, sampling density, density compensation, and reconstruction method. The results also demonstrate that a trajectory that produces a better noiseless reconstruction does not necessarily produce the most noise-resistant reconstruction.

G. Noise and Motion Comparison

In the noise and motion comparison, Cartesian and VDC generally performed the best. The severe image degradation observed for the radial and spiral trajectories between the 40 dB and 30 dB scenarios suggests that noise was the primary contributor to image degradation under the conditions tested.

However, the fact that Cartesian and VDC performed better under noisy conditions does not necessarily mean that they are better overall than radial and spiral trajectories. Noise is an important factor in MRI, but it can potentially be reduced through improved signal processing, noise reduction techniques, or better acquisition conditions without necessarily increasing acquisition time. For this trajectory comparison, motion resistance is considered a higher priority because patient motion is difficult to completely eliminate during an acquisition. As demonstrated in the motion experiments, the choice of sampling trajectory can have a significant effect on the resulting artifacts. Therefore, although Cartesian and VDC showed better noise resistance, radial and spiral may still provide advantages in practical MRI applications where motion resistance and preservation of fine detail are more important.

H. Overall Discussion

A direct comparison of real acquisition times is not possible without access to a real MRI system. It is also important to note that the trajectory used during acquisition is not the only method used to reduce acquisition time. Methods such as parallel imaging, simultaneous acquisition, and artificial intelligence-based reconstruction are also being used and developed [31]. However, the sampling trajectory remains an important part of MRI acquisition, and future improvements will likely involve combining improved trajectories with these other techniques.

Another factor that was not considered in this study is the effect of spin and gradient echoes caused by changing magnetic fields during an MRI acquisition. These effects can influence the measured signal and may introduce additional sources of error that are not represented in the current simulation. This could be particularly important for certain trajectories where the gradient is continuously changing or where the acquisition requires rapid changes in gradient direction [30].

The purpose of this study was not to identify a universally optimal k-space trajectory or determine definitive acquisition parameters. Rather, the objective was to demonstrate the capabilities of the developed simulation program and evaluate its potential as a workflow for investigating and optimizing k-space sampling trajectories. The program can systematically vary trajectory parameters, evaluate their effects on reconstruction quality and computational performance, and

identify promising configurations for further investigation. The program was also designed to be modular, allowing additional trajectories such as golden-angle radial, other variable-density trajectories, and propeller trajectories to be added in the future. The program could also be expanded to support three-dimensional acquisition.

Although the optimal parameters identified in this study are specific to the tested images, sampling conditions, and reconstruction methods, the workflow provides a foundation for more extensive optimization using additional datasets, realistic acquisition constraints, and additional quantitative evaluation methods. Future work should also incorporate physical MRI system limitations, including gradient constraints and acquisition timing, and compare the simulated results with real MR acquisitions. This would help determine how well the trends observed in simulation translate to real MRI systems.

VI. Conclusion

The results of the NUFFT performance comparison showed that the NUFFT algorithm provided a substantial computational efficiency improvement over the original NUDFT, achieving up to approximately 4× faster reconstruction at higher sampling levels while maintaining similar reconstruction quality. This demonstrates that the NUFFT can provide a useful alternative to the direct NUDFT when reconstructing non-Cartesian k-space data.

The trajectory comparisons showed that no single trajectory was the best under every condition tested. Cartesian Variable Density (VDC) generally produced the highest PSNR and SSIM values, particularly at higher sampling percentages. However, radial and spiral trajectories often preserved fine image details better than VDC, especially at lower sampling percentages. Spiral provided a particularly good overall balance between reconstruction quality and preservation of fine details. Radial also showed strong resistance to the simulated motion used in this study.

The noise and motion experiments also demonstrated that trajectory performance depends on the conditions under which the trajectory is used. Cartesian and VDC showed greater resistance to the simulated noise, while radial and spiral

generally showed greater degradation as noise increased. However, the DCF experiment demonstrated that reconstruction methods can also have a significant effect on these results. This shows that trajectory performance cannot necessarily be evaluated independently from the reconstruction method used.

Validation against acquisitions from a physical MRI system was outside the scope of this study. Therefore, the results should be considered specific to the simulated acquisition, reconstruction methods, test images, and conditions used in this study. Real MRI systems also have additional limitations, including gradient amplitude and slew-rate limits, acquisition timing, patient motion, and other hardware constraints that were not directly modeled in this work.

The main purpose of this study was not to identify a universally optimal k-space trajectory or determine definitive acquisition parameters. Instead, the goal was to demonstrate the capabilities of the developed simulation program and evaluate a possible workflow for investigating and optimizing k-space sampling trajectories. The program allows trajectory parameters to be systematically changed and their effects on reconstruction quality and computational performance to be measured. New trajectories, additional reconstruction methods, and three-dimensional sampling patterns could also be added to the program in future work.

Although this study cannot determine which trajectory is best for real MRI systems, it demonstrates that different trajectories have different advantages and drawbacks depending on the sampling conditions and evaluation criteria. The results also show the potential of simulation-based testing as an early step in the development and evaluation of new k-space trajectories before testing them on a physical MRI system.

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